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NDIC - DATEX II Elaborated Data Publication - Travel time

Release 1.1rev0

Národní Dopravní Informační Centrum

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CONTENTS

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This specification defines format, used for publishing travel times of personal cars and lories on set of predefined locations.

CHAPTER

ONE

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format
: description of data structure for transferring some kind of information

protocol
: description of data transfer process and rules

NDIC
: national traffic information centre

DATEX II model
: class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [d2ref]. For additional information see [d2supp].

(DATEX II) profile
: simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension
extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension
Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension
Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication
an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema
language, which allows definition of rules for a XML document structure.

XSD
XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri
uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CHAPTER TWO

CONCEPTS

Described format allows publishing travel times on set of predefined routes.

2.1 Publication type ElaboratedDataPublication

Format is based on DATEX II publication of type *ElaboratedDataPublication* [16157-5] and is used to publish information about travel times on defined routes.

```xml
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
    <feedType>x-format:cz-ndic_d2-travel-time-v1.1</feedType>
    <publicationTime>2017-11-18T00:00:06+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <periodDefault>300</periodDefault>
    <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <elaboratedData>
      <source>
        <sourceIdentification>SectionID=1</sourceIdentification>
      </source>
```

Travel time for particular route is given in element *elaboratedData*, which can appear in the publication multiple times, describing travel times on distinct routes. Element *elaboratedData* does not have its own identifier. Its uniqueness is guaranteed by publication time and by identifier of referenced location.

2.2 Route description (by predefined locations)

The *route* is described by a linear section on the road network, given by the presence of a vehicle detection. Since the position of the vehicle detectors or sections where the time is calculated does not change, the location in this profile is described by a reference (id) to a predefined location set defined in different document. The reference points to a class instance *PredefinedLocation* from package *PredefinedLocationsPublication*. Therefore, the location is not described directly in the file.

Route of status information (travel time) is determined by reference to a set of predefined locations. To look up given location, it is necessary to obtain publication *PredefinedLocationsPublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *PredefinedLocation*), with the same identifier and version.

The route for which the calculated travel time is valid is specified by the *pertinentLocation* element. It always has *xsi:type="LocationByReference"* and includes element *predefinedLocationReference* with the attribute *targetClass="PredefinedLocation"*.

```
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id="56472218-da2f-4986-880a-3ad8e3d2bf32" targetClass="PredefinedLocation" version="1"/>
        </pertinentLocation>
```

2.3 Travel time of personal cars and lories

Travel time is expressed in seconds in attribute *travelTime* of type *DurationValue* in class *TravelTimeData*.

Only current calculated travel time is expressed, other information (free flow travel time, normal travel time, free flow velocity, travel time trend) is not used.

Profile provides only data calculated for personal cars and lorries and vehicles without type specification.

```
        <vehicleType>anyVehicle</vehicleType>
        <travelTime>
          <duration>251</duration>
        </travelTime>
```

2.4 Publishing only currently available data

Format provides travel time only for routes, for which data are available. Other routes, with no data available are omitted.

CHAPTER

THREE

TIME VALIDITY

The start of validity of travel times is given once for all records in the publication by the *timeDefault* element, of the type *DateTime* and class *ElaboratedDataPublication*.

The stated validity time represents the current time to which the calculated data relates, it is the end of the measurement period.

No further specification of validity is used.

```xml
    <feedType>x-format:cz-ndic_d2-travel-time-v1.1</feedType>
    <publicationTime>2017-11-18T00:00:06+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <periodDefault>300</periodDefault>
    <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
```

CHAPTER FOUR

LOCATION REFERENCING

The format describes locations exclusively by reference to a set of predefined routes (class *predefinedLocation*) in publication *PredefinedLocationsPublication* [16157-3]..

Location referencing methods are out of scope of this document and are described elsewhere.

CHAPTER

FIVE

APPENDIX A: MESSAGE EXAMPLES

5.1 travel-time.xml - travel time on a route

The publication contain one record about travel time on one route.

Routes are defined in different publication of type *PredefinedLocationsPublication*, which is not described here in detail.

The route, with identifier *56472218-da2f-4986-880a-3ad8e3d2bf32*, has travel time *251* seconds.

```xml
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
    <feedType>x-format:cz-ndic_d2-travel-time-v1.1</feedType>
    <publicationTime>2017-11-18T00:00:06+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <periodDefault>300</periodDefault>
    <timeDefault>2017-11-17T23:59:35+01:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <elaboratedData>
      <source>
        <sourceIdentification>SectionID=1</sourceIdentification>
      </source>
      <basicData xsi:type="TravelTimeData">
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id="56472218-da2f-4986-880a-3ad8e3d2bf32" targetClass="PredefinedLocation" version="1"/>
        </pertinentLocation>
        <vehicleType>anyVehicle</vehicleType>
        <travelTime>
          <duration>251</duration>
        </travelTime>
```

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```
      </basicData>
    </elaboratedData>
    <elaboratedData>
      <source>
        <sourceIdentification>SectionID=2</sourceIdentification>
      </source>
      <basicData xsi:type="TravelTimeData">
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id=" 26353878-23a3-4937-8933-3d763399a040"␣
↪targetClass="PredefinedLocation" version="1"/>
        </pertinentLocation>
        <vehicleType>anyVehicle</vehicleType>
        <travelTime>
          <duration>250</duration>
        </travelTime>
      </basicData>
    </elaboratedData>
    <elaboratedData>
      <source>
        <sourceIdentification>SectionID=3</sourceIdentification>
      </source>
      <basicData xsi:type="TravelTimeData">
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id="d8063ebf-5add-4562-aaaa-1afe3b5e6065"␣
↪targetClass="PredefinedLocation" version="1"/>
        </pertinentLocation>
        <vehicleType>anyVehicle</vehicleType>
        <travelTime>
          <duration>285</duration>
        </travelTime>
      </basicData>
    </elaboratedData>
  </payloadPublication>
</d2LogicalModel>
```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *elaboratedData*, (in our case there is one).

Attribute *xsi:type="ElaboratedDataPublication"* in element *payloadPublication* determines, that this envelope is of type Elaborated data publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* gives time when publication has been created.

Element *publicationCreator* identifies creator of the publication.

timeDefault element gives start time of validity for all of calculated travel times, the time, for which all travel time values were calculated.

Element *periodDefault* gives the sampling interval in seconds.

5.1.2 Description of calculated travel time on one route

Following element *elaboratedData* can be repeated and describes a calculated travel time on a particular route.

Element *sourceIdentification* gives the identifier to the measuring site / equipment additionally to location identifier.

Nested element *basicData* states by it's attribute *xsi:type="TravelTimeData"*, that given values are of type "travel time".

Travel time location

The attribute *id ="56472218-da2f-4986-880a-3ad8e3d2bf32" in element `predefinedLocationReference* refers to a predefined location (route) with a given identifier, and the attribute *version ='1'* specifies the version of the location description.

Travel time

Element *travelTime* gives the travel time *duration* in seconds, in this case *199* seconds.

Time validity

Time validity (start time) of calculated travel times is determined together for all records in element *timeDefault*.

CHAPTER

SIX

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```xml
<?xml version="1.0" encoding="utf-8" standalone="no"?>
<xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified" xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace="http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="_ExtensionType">
    <xs:sequence>
      <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="_PredefinedLocationVersionedReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:VersionedReference">
        <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="AreaOfInterestEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="continentWide" />
      <xs:enumeration value="national" />
      <xs:enumeration value="neighbouringCountries" />
      <xs:enumeration value="notSpecified" />
      <xs:enumeration value="regional" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="BasicData" abstract="true">
    <xs:sequence>
      <xs:element name="pertinentLocation" type="D2LogicalModel:GroupOfLocations" minOccurs="0" />
      <xs:element name="basicDataExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="Boolean">
    <xs:restriction base="xs:boolean" />
  </xs:simpleType>
  <xs:simpleType name="ComputationMethodEnum">
    <xs:restriction base="xs:string">
```

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```
      <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" />
      <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
      <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
      <xs:enumeration value="medianOfSamplesInATimePeriod" />
      <xs:enumeration value="movingAverageOfSamples" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="ConfidentialityValueEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="internalUse" />
      <xs:enumeration value="noRestriction" />
      <xs:enumeration value="restrictedToAuthorities" />
      <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="CountryEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="at" />
      <xs:enumeration value="be" />
      <xs:enumeration value="bg" />
      <xs:enumeration value="ch" />
      <xs:enumeration value="cs" />
      <xs:enumeration value="cy" />
      <xs:enumeration value="cz" />
      <xs:enumeration value="de" />
      <xs:enumeration value="dk" />
      <xs:enumeration value="ee" />
      <xs:enumeration value="es" />
      <xs:enumeration value="fi" />
      <xs:enumeration value="fo" />
      <xs:enumeration value="fr" />
      <xs:enumeration value="gb" />
      <xs:enumeration value="gg" />
      <xs:enumeration value="gi" />
      <xs:enumeration value="gr" />
      <xs:enumeration value="hr" />
      <xs:enumeration value="hu" />
      <xs:enumeration value="ie" />
      <xs:enumeration value="im" />
      <xs:enumeration value="is" />
      <xs:enumeration value="it" />
      <xs:enumeration value="je" />
      <xs:enumeration value="li" />
      <xs:enumeration value="lt" />
      <xs:enumeration value="lu" />
      <xs:enumeration value="lv" />
      <xs:enumeration value="ma" />
      <xs:enumeration value="mc" />
      <xs:enumeration value="mk" />
      <xs:enumeration value="mt" />
      <xs:enumeration value="nl" />
      <xs:enumeration value="no" />
      <xs:enumeration value="pl" />
```

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```
      <xs:enumeration value="pt" />
      <xs:enumeration value="ro" />
      <xs:enumeration value="se" />
      <xs:enumeration value="si" />
      <xs:enumeration value="sk" />
      <xs:enumeration value="sm" />
      <xs:enumeration value="tr" />
      <xs:enumeration value="va" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel" />
  <xs:complexType name="D2LogicalModel">
    <xs:sequence>
      <xs:element name="exchange" type="D2LogicalModel:Exchange" />
      <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"␣
↪minOccurs="0" />
      <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
  </xs:complexType>
  <xs:complexType name="DataValue" abstract="true">
    <xs:sequence>
      <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
    <xs:attribute name="computationalMethod" type=
↪"D2LogicalModel:ComputationMethodEnum" use="optional" />
    <xs:attribute name="numberOfIncompleteInputs" type=
↪"D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="numberOfInputValuesUsed" type=
↪"D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
    <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional"␣
↪/>
    <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
↪" use="optional" />
  </xs:complexType>
  <xs:simpleType name="DateTime">
    <xs:restriction base="xs:dateTime" />
  </xs:simpleType>
  <xs:complexType name="DurationValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="duration" type="D2LogicalModel:Seconds" minOccurs="1"␣
↪maxOccurs="1" />
          <xs:element name="durationValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
```

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```
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="ElaboratedData">
    <xs:sequence>
      <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
      <xs:element name="basicData" type="D2LogicalModel:BasicData" minOccurs="0" />
      <xs:element name="elaboratedDataExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ElaboratedDataPublication">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:PayloadPublication">
        <xs:sequence>
          <xs:element name="periodDefault" type="D2LogicalModel:Seconds" minOccurs="0" maxOccurs="1" />
          <xs:element name="timeDefault" type="D2LogicalModel:DateTime" minOccurs="0" maxOccurs="1" />
          <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation" />
          <xs:element name="elaboratedData" type="D2LogicalModel:ElaboratedData" maxOccurs="unbounded" />
          <xs:element name="elaboratedDataPublicationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="Exchange">
    <xs:sequence>
      <xs:element name="supplierIdentification" type="D2LogicalModel:InternationalIdentifier" />
      <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="Float">
    <xs:restriction base="xs:float" />
  </xs:simpleType>
  <xs:complexType name="GroupOfLocations" abstract="true">
    <xs:sequence>
      <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="HeaderInformation">
    <xs:sequence>
      <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum" minOccurs="0" maxOccurs="1" />
      <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="headerInformationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
```

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```
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="InformationStatusEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="real" />
      <xs:enumeration value="securityExercise" />
      <xs:enumeration value="technicalExercise" />
      <xs:enumeration value="test" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="InternationalIdentifier">
    <xs:sequence>
      <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"␣
↪maxOccurs="1" />
      <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1
↪" maxOccurs="1" />
      <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="Language">
    <xs:restriction base="xs:language" />
  </xs:simpleType>
  <xs:complexType name="Location" abstract="true">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:GroupOfLocations">
        <xs:sequence>
          <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="LocationByReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:Location">
        <xs:sequence>
          <xs:element name="predefinedLocationReference" type="D2LogicalModel:_
↪PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1" />
          <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MultilingualString">
    <xs:sequence>
      <xs:element name="values">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"␣
↪maxOccurs="unbounded" />
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
```

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```
  </xs:complexType>
  <xs:complexType name="MultilingualStringValue">
    <xs:simpleContent>
      <xs:extension base="D2LogicalModel:MultilingualStringValueType">
        <xs:attribute name="lang" type="xs:language" />
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
  <xs:simpleType name="MultilingualStringValueType">
    <xs:restriction base="xs:string">
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="NonNegativeInteger">
    <xs:restriction base="xs:nonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="PayloadPublication" abstract="true">
    <xs:sequence>
      <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
      <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1" maxOccurs="1" />
      <xs:element name="publicationCreator" type="D2LogicalModel:InternationalIdentifier" />
      <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
  </xs:complexType>
  <xs:simpleType name="Percentage">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:simpleType name="Seconds">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:complexType name="Source">
    <xs:sequence>
      <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
      <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="String">
    <xs:restriction base="xs:string">
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="TravelTimeData">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:BasicData">
        <xs:sequence>
          <xs:element name="travelTimeType" type="D2LogicalModel:TravelTimeTypeEnum" minOccurs="0" maxOccurs="1" />
          <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs="0" maxOccurs="unbounded" />
```

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```
          <xs:element name="travelTime" type="D2LogicalModel:DurationValue" minOccurs=
↪"0" />
          <xs:element name="travelTimeDataExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="TravelTimeTypeEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="best" />
      <xs:enumeration value="estimated" />
      <xs:enumeration value="instantaneous" />
      <xs:enumeration value="reconstituted" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="VehicleTypeEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="anyVehicle" />
      <xs:enumeration value="car" />
      <xs:enumeration value="lorry" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="VersionedReference">
    <xs:attribute name="id" type="xs:string" use="required" />
    <xs:attribute name="version" type="xs:string" use="required" />
  </xs:complexType>
</xs:schema>
```

CHAPTER

SEVEN

CHANGELOG

version 1.1.0

- added feed type to schema and samples
- added sourceIdentification to samples
- added lories, anyVehicle, removed vans from vehicle spec class, documentation changed accordingly.

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.0.0

- Initial format and documentation

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